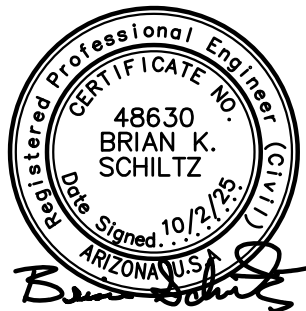


**Preliminary Basis of Design Report
Waste Water
For
Finish Line Auto Club – Perimeter Center
Princess Drive and Loop 101
Scottsdale, Arizona 85255**

**COS Plan Check #: 1059-PA-2024
COS Case #: 19-ZN-2004#2**

**1st Submittal 5/8/25
2nd Submittal 8/11/25
3rd Submittal 10/2/25**



**Prepared by:
Brian Schiltz
Hunter Engineering, Inc.
10446 N. 74th Street. Suite 140
Scottsdale, AZ 85258**

H.E. PROJECT NO. LGEC332

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1.0 Introduction

This design report has been prepared under a contract from LGE Corporation, developer of the Finish Line Auto Club project. The purpose of this report is to provide a sewer analysis, required by the City of Scottsdale, to support this development. Preparation of this report has been done according to the procedures detailed in Chapter 7 of the City of Scottsdale's *Design Standards and Policy Manual* (Reference 1).

This development project is located along the west side of N. Perimeter Drive and along the east side of N. Pacesetter Way, between E. Pacesetter Way and E. Princess Drive, within the City of Scottsdale, Maricopa County, Arizona. The site is specifically located in the south half of Section 36, Township 4 North, Range 4 East of the Gila and Salt River Base and Meridian. Figure 1, in Appendix A, illustrates the location of the project site in relation to the City of Scottsdale street system. Access to the site is provided from N. Perimeter Drive and N. Pacesetter Way.

The proposed project site is currently a developed parcel within the Perimeter Center master development. The existing parcel is bound by existing commercial developments to the north and south, N. Perimeter Drive to the east and N. Pacesetter Way to the west.

The development proposes the construction of four new car condominium units, denoted as Buildings B through E, and a car condominium Clubhouse, denoted as Building A, on the easterly side of the project. This campus is dedicated to a private auto club. A car storage building is proposed on the westerly side of the parcel. This building totals 49,266 square feet and will be classified as a flex-industrial building. Site improvements will include construction of driveway entrances, parking lots, sidewalk/hardscape, landscape areas, and supporting infrastructure including new storm water drainage systems, water, sewer and fire line services. The overall project site is approximately 10.39± acres.

2.0 Existing Condition

The existing site consists of a single commercial building and associated parking lots with parking canopies. The site is part of the overall Scottsdale Perimeter Center master development. There are existing 8-inch VCP public sewer mains within N. Perimeter Drive and N. Pacesetter Way adjacent to the site. There are four existing 8-inch sewer services to the site, two off N. Perimeter Drive and two off N. Pacesetter Way.

The City of Scottsdale Water and Wastewater Department requested that three downstream sewer manholes be flow tested to determine the available capacity. The first manhole tested (MH1) is located at East Bell Road east of North Hayden Road. The flow test was performed by RDH Environmental Services, LLC between October 31st, 2024 and November 11th, 2024. MH1 near Bell and Hayden resulted in a highest measured maximum flow during the test period of 184.97 gpm. The main tested is a 15-inch PVC pipe. The capacity for this main is 1,152 gpm at a minimum 0.225% slope and a d/D ratio of 0.70. This existing main is at approximately 16% capacity with approximately 967 gpm of available flow. It should be noted that the city quarter section maps show this existing main as 12-inch, however the main was measured as 15-inch in the field. Those measurements are shown in the flow test report in Appendix C.

The second manhole tested is located at East Bell Road east of North Hayden Road. The flow test was performed by RDH Environmental Services, LLC between September 19th, 2025 and September 29th, 2025. This manhole resulted in a highest measured maximum flow during the test period of 204.60 gpm. The main tested is a 15-inch PVC pipe. Per the flow test analysis, the estimated total maximum flow is 752 gpm with an 80% maximum of 601 gpm. Using the 80% maximum total flow estimate, this existing main is at approximately 34% capacity with approximately 396 gpm of available flow. Those measurements are shown in the flow test report in Appendix C.

The third manhole tested is located at North Hayden Road, north of East Bell Road. The flow test was performed by RDH Environmental Services, LLC between September 19th, 2025 and September 29th, 2025. This manhole resulted in a highest measured maximum flow during the test period of 153.93 gpm. The main tested is an 8-inch PVC pipe. Per the flow test analysis, the estimated total maximum flow is 377 gpm with an 80% maximum of 301 gpm. Using the 80% maximum total flow estimate, this existing main is at approximately 51% capacity with approximately 147 gpm of available flow. Those measurements are shown in the flow test report in Appendix C.

3.0 Proposed Sanitary Sewer System

This project proposes to connect to the two existing 8-inch sewer services off N. Perimeter Drive and one of the existing 8-inch sewer service off N. Pacesetter Way. The proposed sewer system for Buildings A, B, C, and D is comprised of 6-inch sewer service lines to each building. Those services flow to a new 8-inch private sewer main with three new manholes. The private sewer main drains southeasterly to the existing 8-inch sewer service and to the 8-inch main in N. Perimeter Drive. The proposed sewer system for Building E is comprised of a 6-inch service line that drains to a proposed manhole located at the northeast corner of the site. This system drains to the existing 8-inch sewer service and to the 8-inch main in N. Perimeter Drive.

The proposed sewer system for the Car Storage Building is comprised of a new 6-inch service that drains to a new monitoring manhole. This monitoring manhole drains to the existing southwesterly 8-inch sewer service and to the 8-inch main in N. Pacesetter Way. Additionally, the car storage building will include a sand/oil separator located at the northeast corner of the building to capture any contaminants from entering the public sewer system. The Conceptual Utility Plan, in Appendix A, depicts the proposed sanitary sewer improvements.

4.0 Projected System Demands

Per Figure 7-1.2 Average Day Sewer Demand in the City of Scottsdale *Design Standards & Policies Manual* a demand of 0.5 gallons per day per building square foot was used for the four car condominium units and the Clubhouse. The owner has constructed several of these types of facilities. From past experience, roughly 50-percent of the condominium garages are expected to add plumbing fixtures and roughly 90-percent of those fixtures are included in singular bathrooms. Additionally, more than 75-percent of the potential plumbing fixtures within the garages will be used less than once per week. As part of the Clubhouse construction, this building is expected to install 16 water closets and 20 sinks for a total of 104 fixture units per Table 709.1 of the 2018 International Plumbing Code. The design peaking factor is 3.0. See Appendix B for a summary of the demand calculations.

Since the City of Scottsdale does not have demands or peaking factors for a car storage building or flex-industrial building, wastewater flows for the car storage building were calculated in accordance with the CSDSPM (Reference 1) and City of Phoenix Water Services Department, Design Standards Manual for Water and Wastewater Systems, 2021. See Appendix C. This results in wastewater flows of 50 gpd per every 1,000 sf of building. As part of this building construction, this building is expected to install 15 water closets and 13 sinks for a total of 86 fixture units per Table 709.1 of the 2018 International Plumbing Code. The peaking factor was calculated using Harmon's Formula per the City of Phoenix design standards. A peaking factor analysis was conducted using two different population estimations. An average provided by the architect assumes a population of one person per 2,100 sf of building and results in a peaking factor of 4.37. The City of Scottsdale zoning parking requirements for manufacturing and industrial use assumes a population of one person per 500 sf of building and results in a peaking factor of 4.24. The greater peaking factor, 4.37, was utilized to determine peak flow. See Appendix B for a summary of these calculations.

According to the calculations provided in Appendix B the proposed site will have an estimated Average Daily Flow total of 45.2 gpm and a Peak Hour Flow of 137.9 gpm. Each building will have its own service with the highest building peak flow of 65.7 gpm. The results of this report are based on the worst-case scenario, with a minimum slope of 1.20% for a 6-inch sewer service for each building.

The capacity of the proposed 6-inch sewer service lines at the minimum slope is 209 gpm on a max d/D ratio of 0.65. This capacity is greater than the Peak Flow of 65.7 gpm for the largest building. Therefore, the maximum individual building flow will be less than the service capacity flow.

The existing 8-inch sewer service stubs to the site were each installed with a slope of 0.34%. Therefore, the capacity of the existing sewer services to the site were calculated based on the minimum slope of 0.34% for an 8-inch sewer line. The existing 8-inch sewer service that connects to the proposed 8-inch private sewer main will drain the highest peak flow totaling 111.3 gpm from Buildings A, B, C, and D. The capacity of the existing 8-inch private sewer service with a minimum slope of 0.34% is 239 gpm with a max d/D ratio of 0.65. This capacity is greater than the total Peak Hour Flow of 111.3 gpm.

The flow tests performed by RDH found a peak flow of 204.60 gpm for the 15-inch sewer main in Bell Road, and 153.93 gpm for the 8-inch sewer main in Hayden Road. The total estimated peak flow, including the proposed development, is 496.43 gpm. The existing 12-inch sewer line that receives these two flows is presumed to have a minimum full flow velocity of 2.5-fps as required by the 2018 City of Scottsdale DSPM, resulting in a minimum slope of 0.3%. Using this worst-case data, the downstream existing 12-inch sewer main, with a max d/D ratio of 0.65, is estimated to flow 662 gpm. Therefore, this 12-inch existing sewer main has a greater capacity than the aforementioned 496.43 gpm. See Appendix B for the pipe calculations.

The sanitary sewer pipe and fitting material for this project has been designated as PVC SDR-35. Trenching and bedding details for this project are to be per MAG Standard Specifications Section 601. Trench width above the installed pipe may be as wide as necessary to properly brace/install the work. Bedding backfill and compaction shall be installed per MAG Standard Specification 601.4. Service lines should connect to sewer according to MAG Standard Detail No. 440-3.

5.0 Conclusion

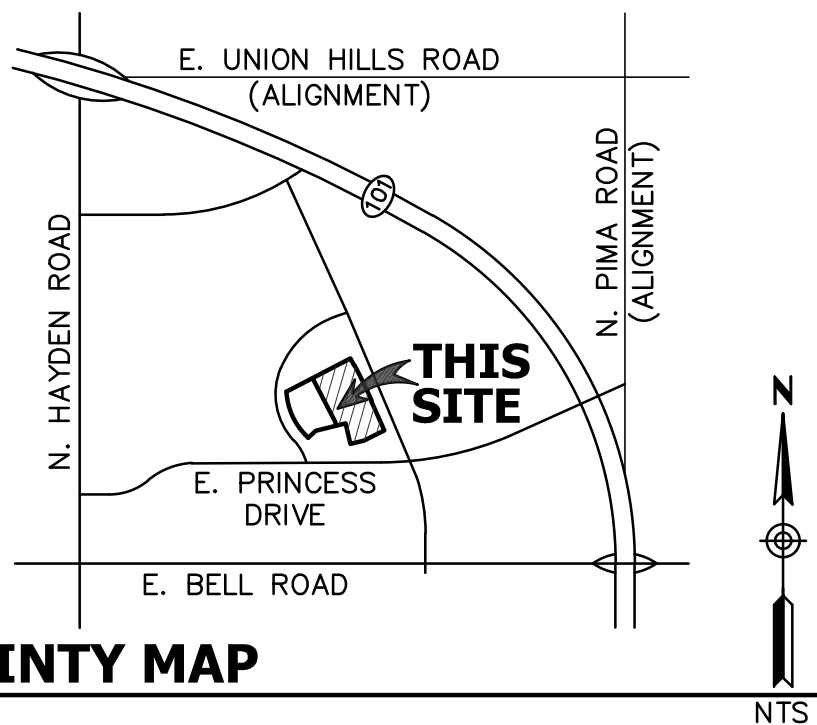
Based on the results of this study, it can be concluded that the proposed sanitary sewer system is adequate to support this development.

6.0 References

- 1) City of Scottsdale Design Standard & Policies Manual, January 2018 (Ref 1).

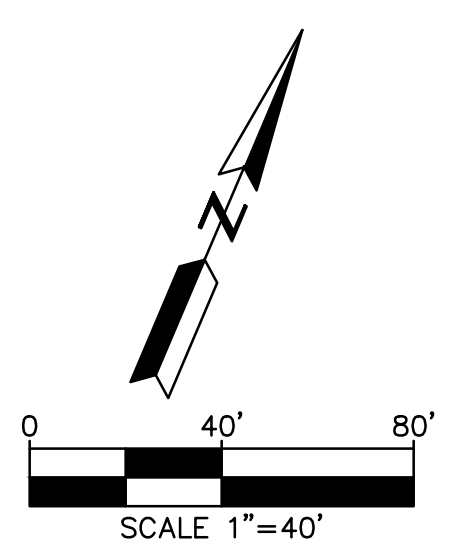
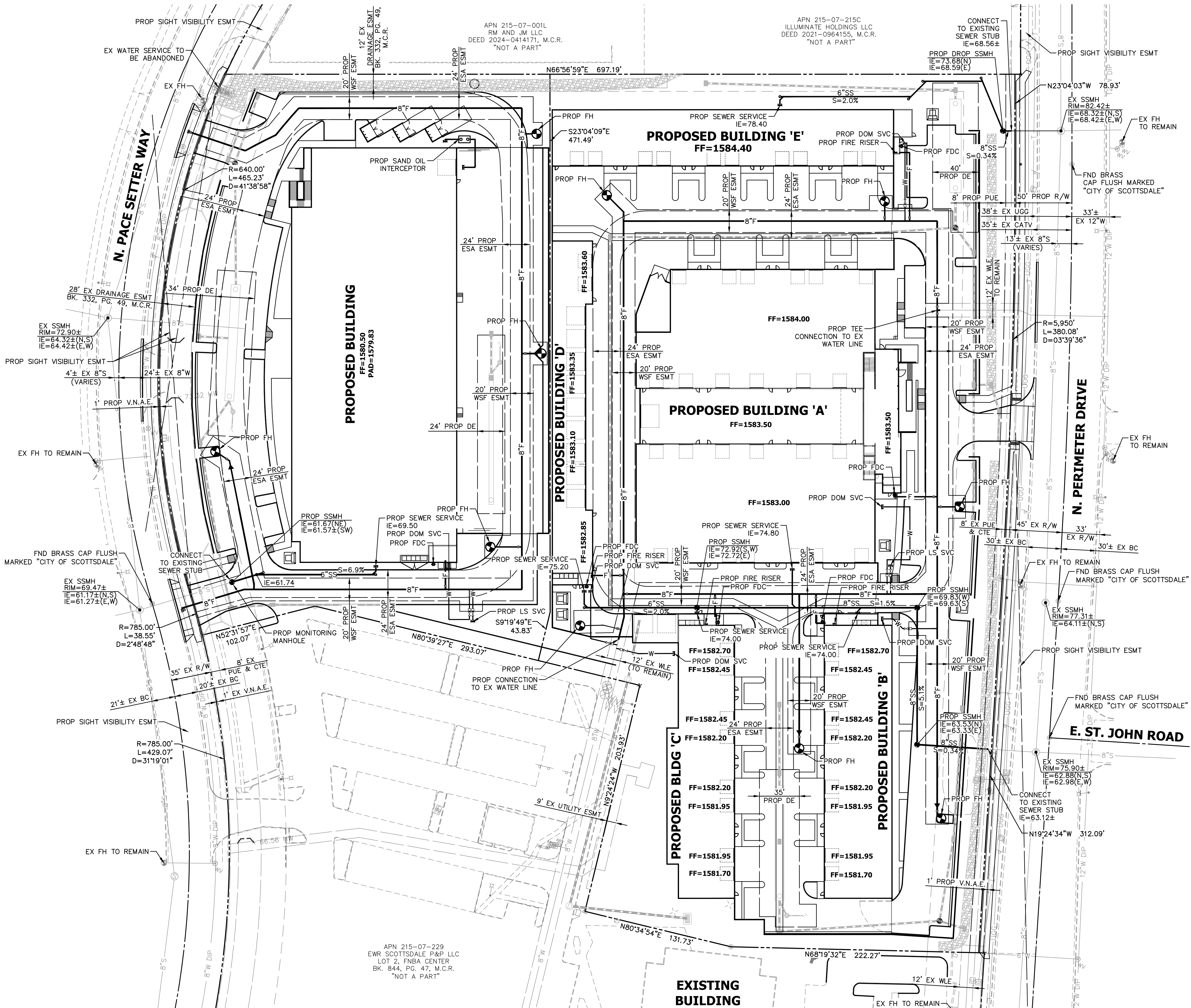
Appendix A

Vicinity Map and Conceptual Utility Plan



VICINTY MAP

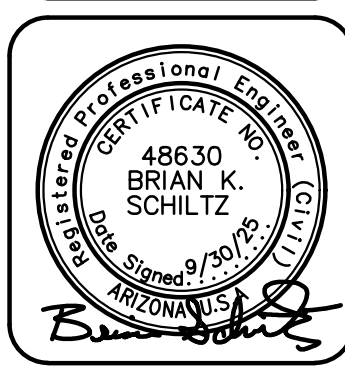
**VICINITY MAP
FIGURE 1**



NO.	DATE	REVISION	BY

DESIGN BY: BKS
DRAWN BY: MM
CHECKED BY: BKS

CIVIL AND SURVEY
HUNTER
ENGINEERING
10446 NORTH 74TH STREET
SUITE 140
SCOTTSDALE, AZ 85258
P 480 991 3985



**CONCEPTUAL UTILITY PLAN
FOR
FINISH LINE AUTO CLUB GARAGES
PRINCESS DRIVE AND LOOP 101
SCOTTSDALE, ARIZONA**



THESE PLANS ARE
NOT APPROVED FOR
CONSTRUCTION
WITHOUT AN
APPROVED SIGNATURE
FROM THE GOVERNING
MUNICIPALITY.

PROJECT NAME:
FINISH LINE
AUTO CLUB
GARAGES

HE NO.: LGEC332
SCALE: 1"=40'

SHEET:
C3

Appendix B
Design Data and Calculations

Project: Finish Line Auto Club - Perimeter Center
Project No.: LGEC332
City: Scottsdale
Date: 9/29/2025

PROJECTED SANITARY SEWER LOADS

Building	Land Use	Building Area (sf)	Average Day Sewer Demand (gpd) Per Table 7-1.2 Average Day Sewer Demand	Average Daily Flow (gpd)	Average Daily Flow (gpm)	Peaking Factor	Peak Flow (gpm)
Building A (Clubhouse)	Commercial	63,020	0.5 gpd/sf	31,510	21.9	3.00	65.7
Building B	Commercial	17,877	0.5 gpd/sf	8,939	6.2	3.00	18.6
Building C	Commercial	17,337	0.5 gpd/sf	8,669	6.0	3.00	18.0
Building D	Commercial	8,712	0.5 gpd/sf	4,356	3.0	3.00	9.0
Subtotal		106,946		53,473	37.1		111.3

Building	Land Use	Building Area (sf)	Average Day Sewer Demand (gpd) Per Table 7-1.2 Average Day Sewer Demand	Average Daily Flow (gpd)	Average Daily Flow (gpm)	Peaking Factor	Peak Flow (gpm)
Building E	Commercial	18,558	0.5 gpd/sf	9,279	6.4	3.00	19.2
Subtotal		18,558		9,279	6.4		19.2

Total (Perimeter Drive)		125,504		62,752	43.5		130.5
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Building	Land Use	Building Area (sf)	Average Day Sewer Demand (gpd) City of Phoenix Water Services Dept. Design Standards Manual Table 8. Water and Wastewater Design Flows	Average Daily Flow (gpd)	Average Daily Flow (gpm)	Peaking Factor Avg provided by Arch: one person per 2,100 sf of building	Peak Flow (gpm)	Peaking Factor COS Zoning Parking Requirement: one person per 500 sf of building	Peak Flow (gpm)
Car Storage Building	Industrial	49,266	50.0 gpd/1000sf	2,463	1.7	4.37	7.4	4.25	7.2
Total (Pacesetter Way)		49,266		2,463	1.7		7.4		

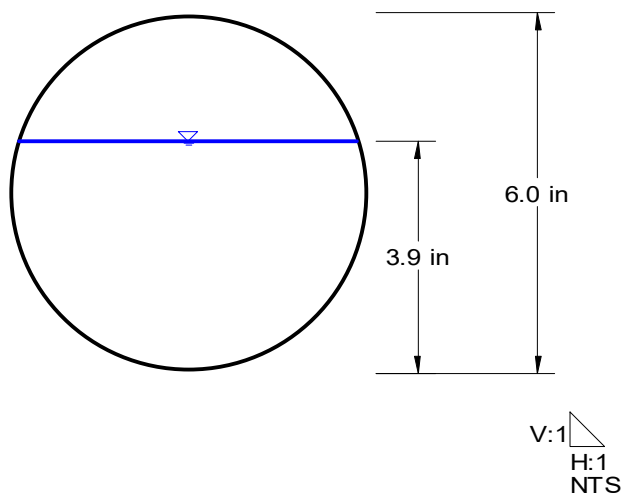
Grand Total		174,770		65,215	45.2		137.9		
--------------------	--	----------------	--	---------------	-------------	--	--------------	--	--

6-inch SS d/D=0.65 S=1.20% **Worksheet for Circular Channel**

Project Description	
Worksheet	6-inch SS d/D=0.65 S=1.20%
Flow Element	Circular Channel
Method	Manning's Formula
Solve For	Discharge

Input Data	
Mannings Coefficient	0.013
Channel Slope	1.20 %
Depth	3.9 in
Diameter	6.0 in

Results	
Discharge	209 gpm
Flow Area	0.1 ft ²
Wetted Perimeter	0.94 ft
Top Width	0.00 ft
Critical Depth	0.35 ft
Percent Full	65.0 %
Critical Slope	1.00 %
Velocity	3.44 ft/s
Velocity Head	0.18 ft
Specific Energy	6.1 in
Froude Number	1.14
Maximum Discharge	297 gpm
Discharge Full	276 gpm
Slope Full	0.69 %
Flow Type	Supercritical

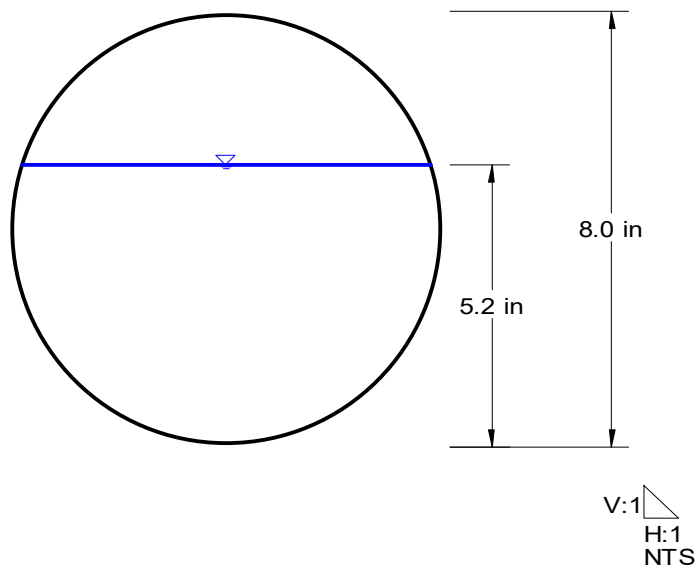


8-inch SS d/D=0.65 S=0.34% Worksheet for Circular Channel

Project Description	
Worksheet	8-inch SS d/D=0.65 S=0.34%
Flow Element	Circular Channel
Method	Manning's Formula
Solve For	Discharge

Input Data	
Mannings Coefficient	0.013
Channel Slope	0.34 %
Depth	5.2 in
Diameter	8.0 in

Results	
Discharge	239 gpm
Flow Area	0.2 ft ²
Wetted Perimeter	1.25 ft
Top Width	0.00 ft
Critical Depth	0.34 ft
Percent Full	65.0 %
Critical Slope	0.71 %
Velocity	2.22 ft/s
Velocity Head	0.08 ft
Specific Energy	6.1 in
Froude Number	0.64
Maximum Discharge	340 gpm
Discharge Full	316 gpm
Slope Full	0.19 %
Flow Type	Subcritical

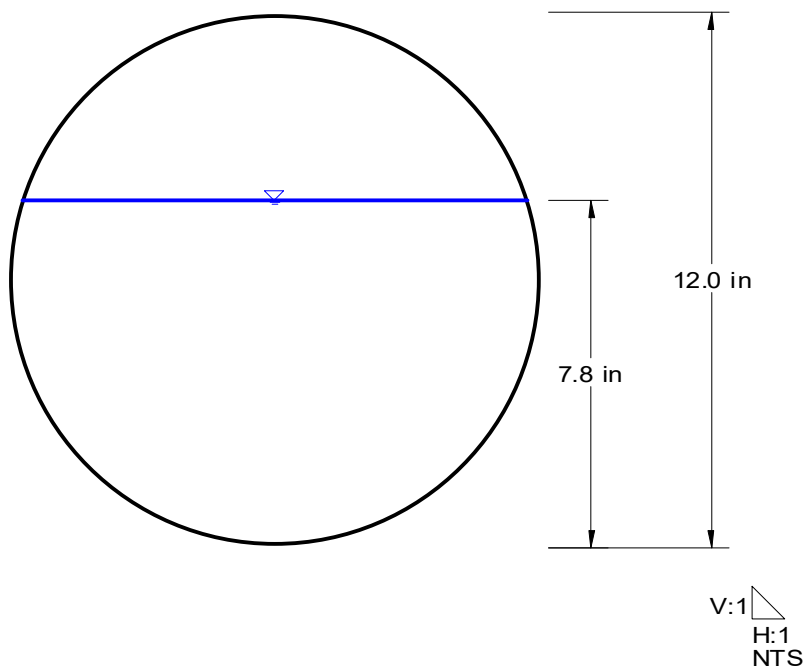


Ex 12-inch SS d/D=0.65 v=2.50 fps Min. Worksheet for Circular Channel

Project Description	
Worksheet	Ex 12-inch SS d/D=0.65 v=2.50 fps Min.
Flow Element	Circular Channel
Method	Manning's Formula
Solve For	Discharge

Input Data	
Mannings Coefficient	0.013
Channel Slope	0.30 %
Depth	7.8 in
Diameter	12.0 in

Results	
Discharge	662 gpm
Flow Area	0.5 ft ²
Wetted Perimeter	1.88 ft
Top Width	0.00 ft
Critical Depth	0.51 ft
Percent Full	65.0 %
Critical Slope	0.62 %
Velocity	2.73 ft/s
Velocity Head	0.12 ft
Specific Energy	9.2 in
Froude Number	0.64
Maximum Discharge	942 gpm
Discharge Full	876 gpm
Slope Full	0.17 %
Flow Type	Subcritical



Appendix C
Reference Information



SL1780 RDH Flow Study for LGE

Blake Wells

LGE Design Build

1200 N. 52nd St, Phoenix, AZ 85008

SL1780 RDH Flow Study, 2 sites total in Scottsdale, AZ from Thursday 10-31-24 to Monday 11-11-24.

Equipment for Site: Hach 901 Logger with Flo-Dar Sensor (Area Velocity).

The equipment was installed on Thursday, 10/31/24 with confined space entry, pipe size confirmed, sensor calibrated, and level depth confirmed to the flow level.

Duration of monitoring: 9 days

Monitor: Flow (gpm), Level (in), and Velocity (fps)

Data logging: 5-minute intervals (No averaged intervals)

Calibration Performed: Calibration method using 3.50-inch target.

Target Measure: 3.50 in Meter Read: 3.53 in 10/31/2024 10:25 am

Meter Validation: PASSED

MH1 located at E. Bell Rd. East of N. Hayden Rd.

72" Diameter, Rim to Invert: 144.00 inches

15" PVC pipe, flowing West

One 4" Lateral pipe with no flow.

The pipe condition is intact with a thin layer of scum on the walls.

Scum line of 4.00 inches

Flo-Dar installed pointing upstream in the 15" pipe channel.

Flow Data is valid having no missing, erroneous, or anomalies with data.

Attached is a MS Excel summary showing level, velocity, and flow logged at 5-minute intervals during the monitoring period.

RDH Environmental Services

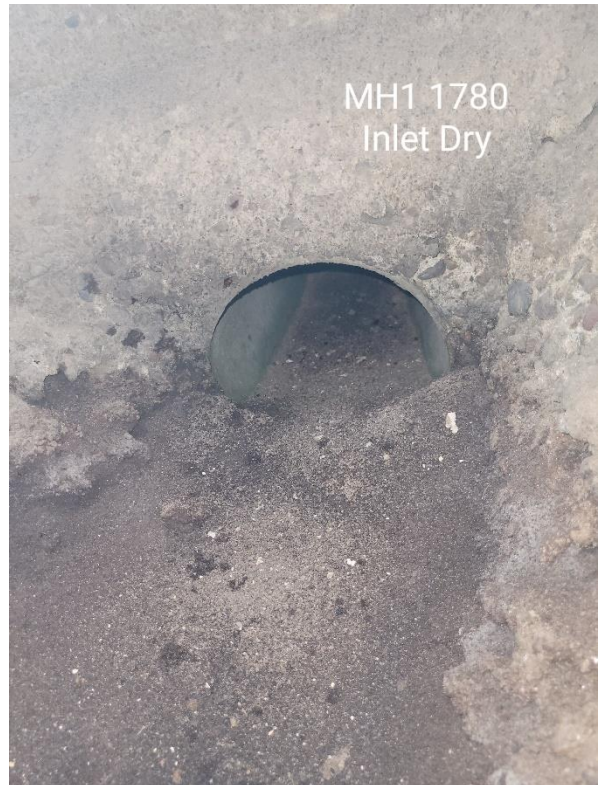
Jeff Schulte

Operations Manager

servicemanager@rdh-env.com

SL1780 RDH Flow Study for LGE

Pictures:





SL1780 RDH Flow Study for LGE

Period Summaries:

LGE COS MH1 Period Summary: Flow				
Measures	Value	Unit	Date	Time
Max.	184.97	gpm	Tuesday, November 5, 2024	2:55 PM
Min.	12.58	gpm	Wednesday, November 6, 2024	3:25 AM
Avg.	63.29	gpm		
Total	977,807.00	gal		

LGE COS MH1 Period Summary: Level				
Measures	Value	Unit	Date	Time
Max.	4.06	in	Thursday, October 31, 2024	11:00 AM
Min.	1.51	in	Wednesday, November 6, 2024	4:55 AM
Avg.	2.48	in		

LGE COS MH1 Period Summary: Velocity				
Measures	Value	Unit	Date	Time
Max.	1.60	fps	Tuesday, November 5, 2024	2:50 PM
Min.	0.42	fps	Wednesday, November 6, 2024	3:25 AM
Avg.	0.97	fps		

*Data begins at 10:30 am on October 31st and ends at 4:00 am on November 11th.



SL1780 RDH Flow Study for LGE

Site Map:



CONFINED SPACE ENTRY PERMIT

ALL COPIES OF PERMIT WILL REMAIN AT JOB SITE UNTIL JOB IS COMPLETED

LOCATION/DESCRIPTION OF CONFINED SPACE

M141 AH 1780

PURPOSE OF ENTRY

Flam Study Installation

EXPECTED HAZARDOUS

Gases

COMMUNICATIONS

Hand & Verbal

ENTRY SUPERVISOR

Jordan Astemborski

DATE

10/31/24

TIME

9:45am

EXPIRATION

12:00pm 11-31-24

SPECIAL REQUIREMENTS BEFORE ENTRY:

	YES	NO		YES	NO
Lockout De-energize - Test and Verify		☒	Escape Harness Required	☒	
Lines Broken - Capped or Blanked		☒	Tripod Emergency Escape Unit	☒	
Purge - Flush and Vent		☒	Lifelines		☒
Ventilation		☒	Fire Extinguishers		☒
Secure Area (Post and Flag)	☒		Lighting (Explosion proof)	☒	
Breathing Apparatus		☒	Protective Clothing	☒	
Resuscitator - Inhalator		☒	Respirator		☒

TEST INTERVAL 15 Min

TEST(S) TO BE TAKEN / ACCEPTABLE ENTRY CONDITIONS

DO NOT ENTER IF PERMISSIBLE ENTRY LEVELS
ARE EXCEEDED

	Permissible Entry Level	DATE	TESTER	TIME	AM/PM	10/31	10/31	10/31								
						4:55	10:10	10:25								
% of Oxygen	19.5% to 23.5%					20.9	20.9	20.9								
^ of L.F.L.* (Gas/Vapor/Mist)	Less than 10%					0	0	0								
Carbon Monoxide	35 ppm (8 hr.)					0	0	0								
Aromatic Hydrocarbon	1 ppm (8 hr.)					0	0	0								
Hydrogen Sulfide	10 ppm (8 hr.)					0	0	0								
Sulfur Dioxide	2 ppm (8 hr.)					0	0	0								
Ammonia	25 ppm (8 hr.)					0	0	0								

NAME OF GAS TESTER(S)

NOTE: Continuous/periodic tests shall be established before beginning the job.

Any questions pertaining to test requirements should be directed to

TESTING INSTRUMENTS USED

Honeywell

NAME

BW Tech

TYPE

GasAlertMax XT II

IDENTIFICATION NUMBER

XT-XWHM-Y-NA MA215-026608

AUTHORIZED ENTRANTS

Zac Schulte

AUTHORIZED ATTENDANTS

Jordan Astemborski

Eric Gentile

PERMIT AUTHORIZATION

I certify that all actions and conditions necessary for safe entry have been performed

Jordan Astemborski

NAME (Print)

Signature

10/31/24

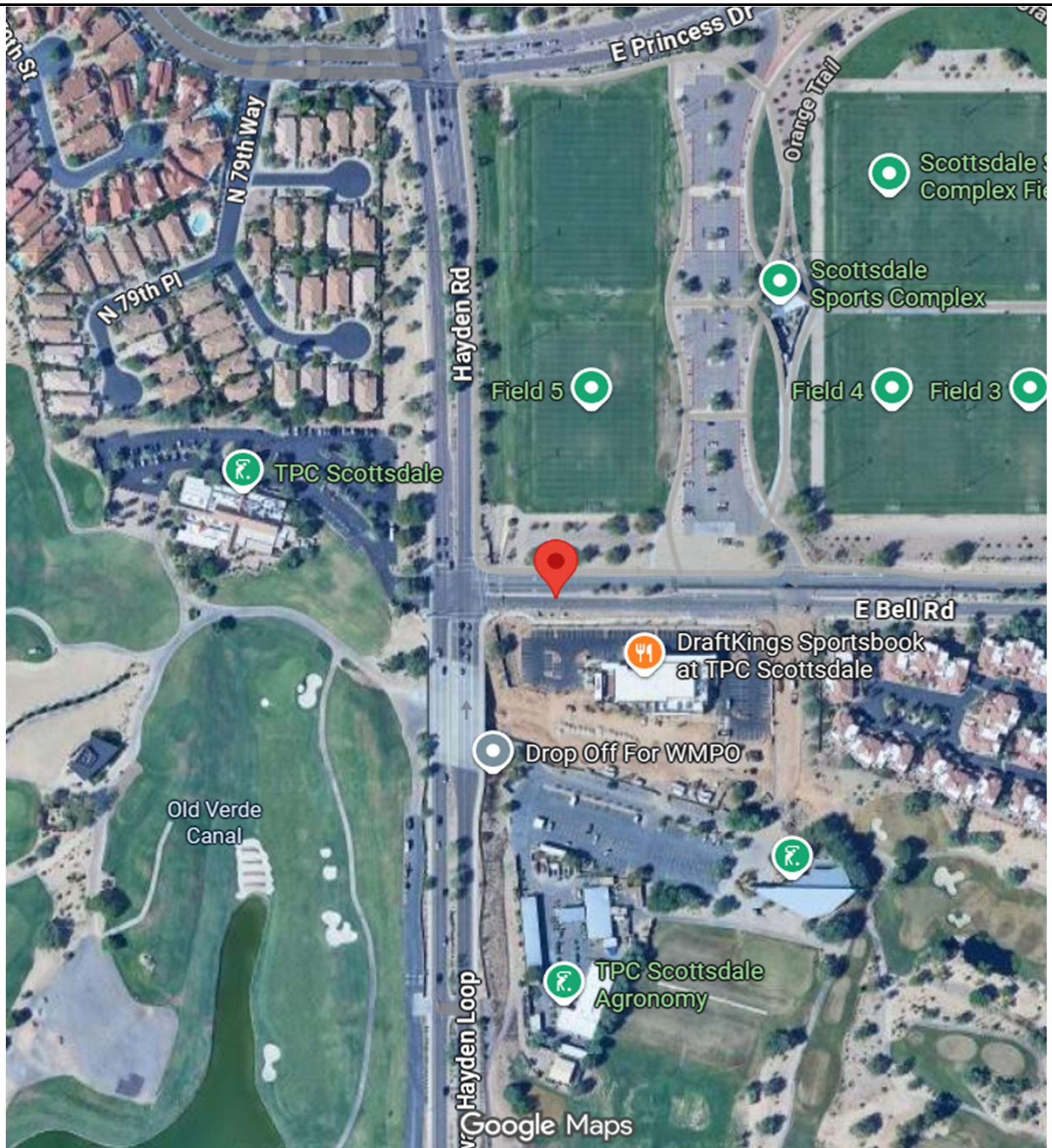
DATE

10:30am

TIME

IN CASE OF AN EMERGENCY CALL 911

RDH Environmental Services (602) 381-1960



Scottsdale Two Site Flow Study

Manhole: N/A
Pipe Size: 14"
Latitude: 33° 38' 24.6" N
Longitude: 111° 54' 24.8" W

Flow Meter Site Report

Meter Information

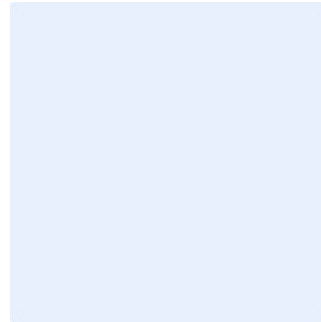
Study Name	Manhole ID	Site Address
LGE Scottsdale	N/A	8129 E Bell Rd, Scottsdale, AZ 85255
Latitude	Longitude	Area Type
33° 38' 24.6" N	111° 54' 24.8" W	Residential
Installation Date	Installation Crew	Sensor Configuration
9/19/2025	Jordan/Zach/Garrick	Velocity & Ultrasonic
Surcharge Mount	Comments	
No	-	



1. Location Map



2. Area Photo



3. Interior MH Photo



4. In Pipe Photo

Manhole Information

Manhole Depth (Feet):	7' 6"
Manhole Material:	PVC

Hydraulics

Surcharge (Feet):	-
Flow Depth at Install (Inches):	2.47
Velocity at Install (FPS):	1.17
Silt Level at Install (Inches):	0
Signs Of Inflow/Infiltration:	No
Overall Hydraulic Rating:	Smooth/Laminar

Pipe Information

Pipe Height (Inches):	12"
Pipe Width (Inches):	-
Pipe Shape:	Circular
Pipe Material:	Lined
Comments:	n/a

Groundwater Gauge

Existing GWG:	No
Installed New GWG:	No
Type:	-
Groundwater Level	
Distance from Gauge to	
Lowest Pipe Invert (Inches):	-

Other Information

Safety Concerns:	-	Required Traffic Control:	No
Required Police Detail:	No	Cell Reception:	Yes
		Comments:	-

Overall Analysis (Wet and Dry Days)

Overall Data

Flow (GPM)

Max	204.60	Average	85.66
Min	30.51		

Level (in)

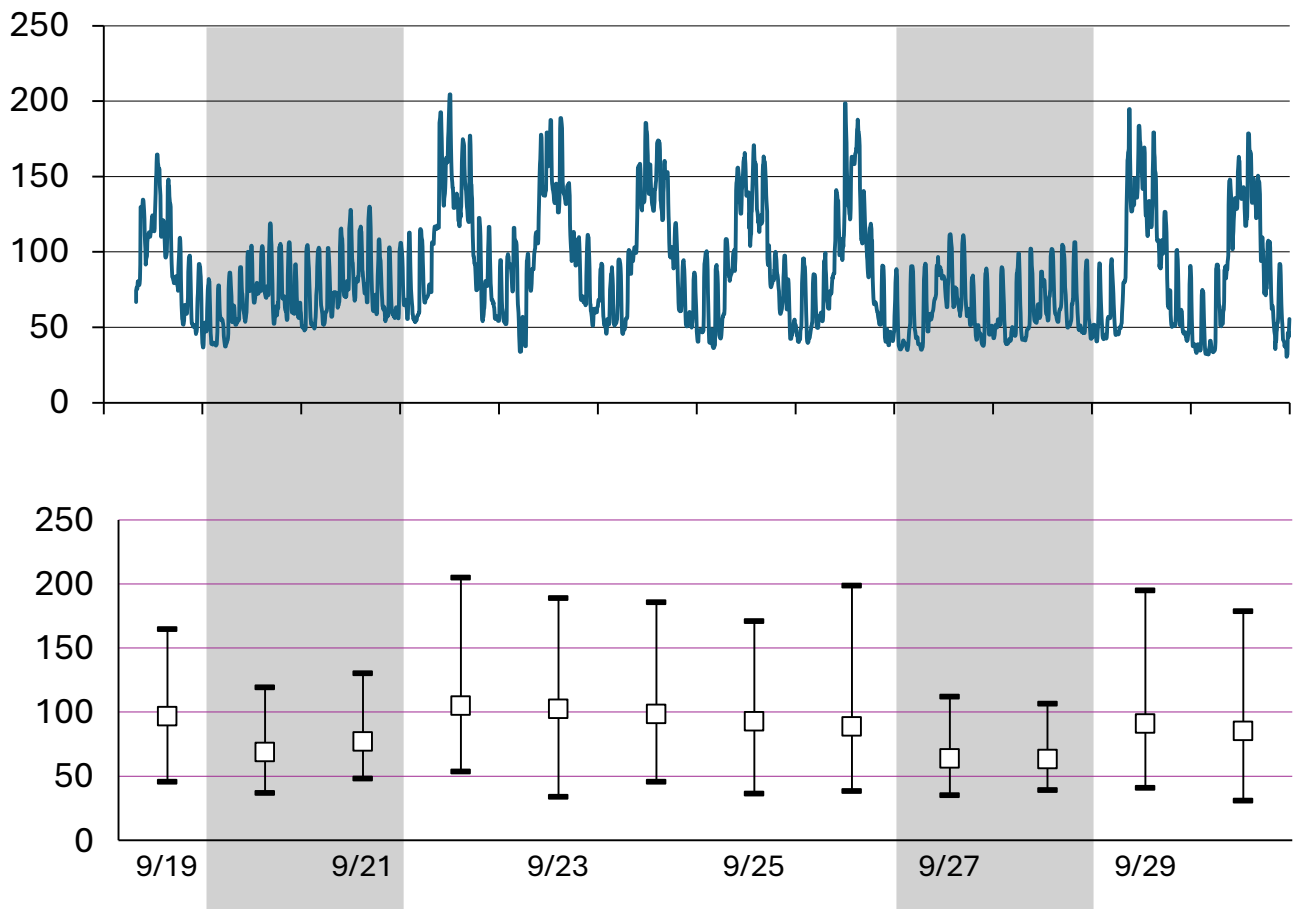
Max	4.35	Average	2.86
Min	1.79		

Velocity (FPS)

Max	1.67	Average	1.16
Min	0.73		

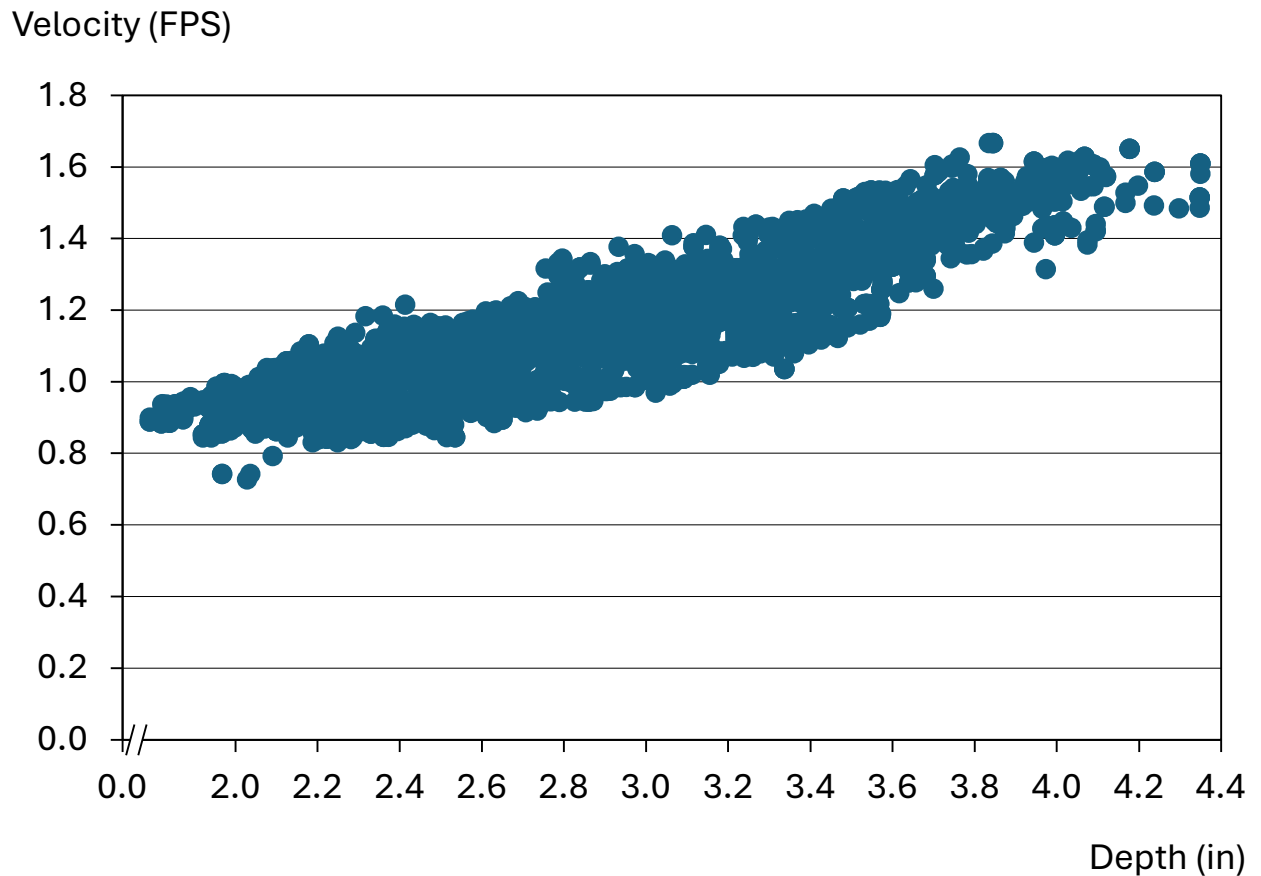
Flow vs. Time

Flow (GPM)



Overall Analysis (Wet and Dry Days)

Velocity vs. Depth



Analysis and notes

Equipment was installed on 9/19. There were three days of rain (9/19, 9/26-27).

Normal diurnal pattern was observed during the work week. Significantly different pattern is observed on the weekends with limited overall flow with no large diurnal pattern.

We used the data of level, flow, and velocity to fit a slope and roughness coefficient consistent. Based off of these input variables, we estimate total maximum flow to be 752 GPM with an 80% maximum to be 601 GPM.

Wet Weather Analysis (9/19, 9/26-9/27)

Overall Data

Flow (GPM)

Max	198.48	Average	81.30
Min	34.90		

Level (in)

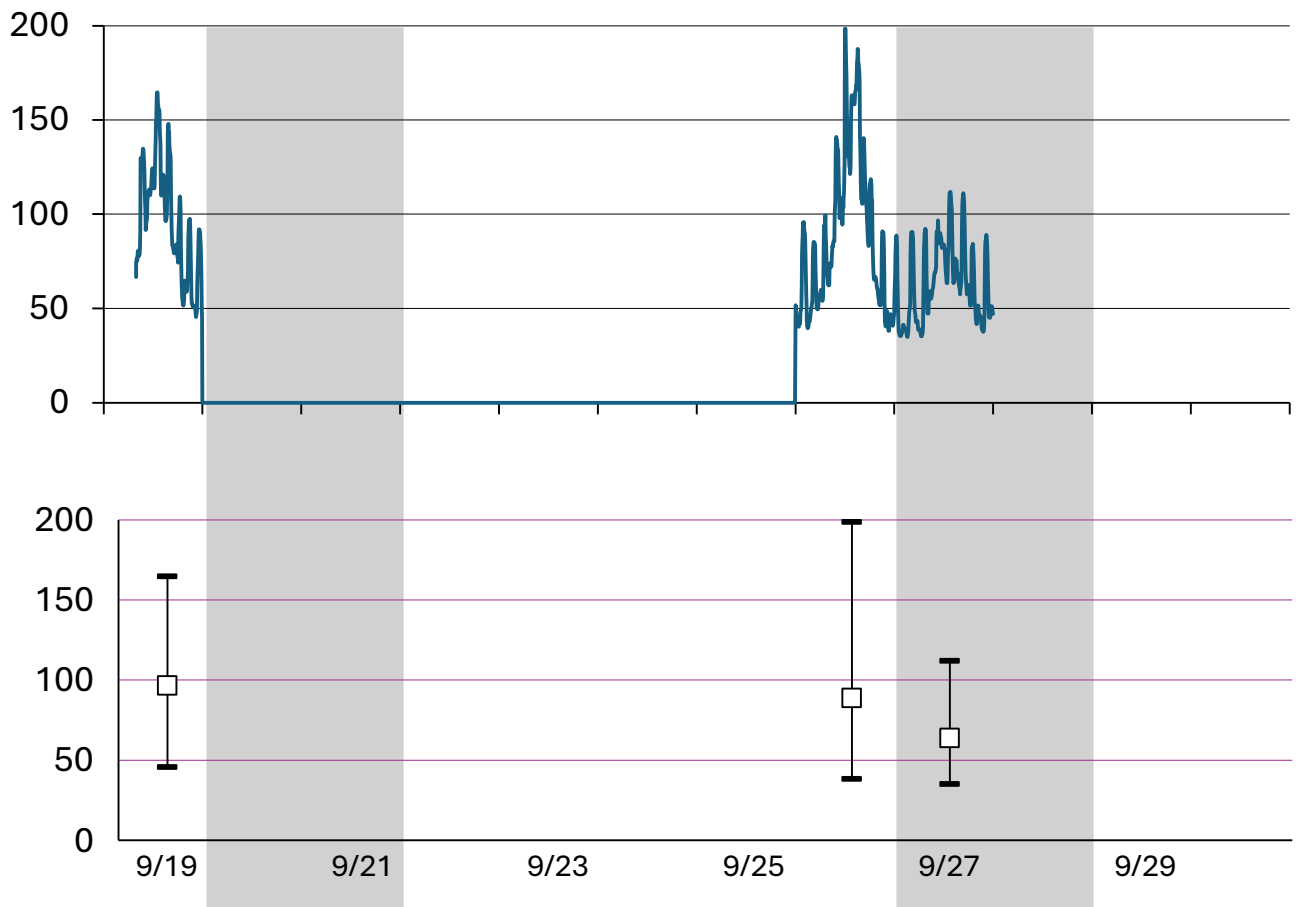
Max	4.18	Average	2.78
Min	1.94		

Velocity (FPS)

Max	1.65	Average	1.15
Min	0.86		

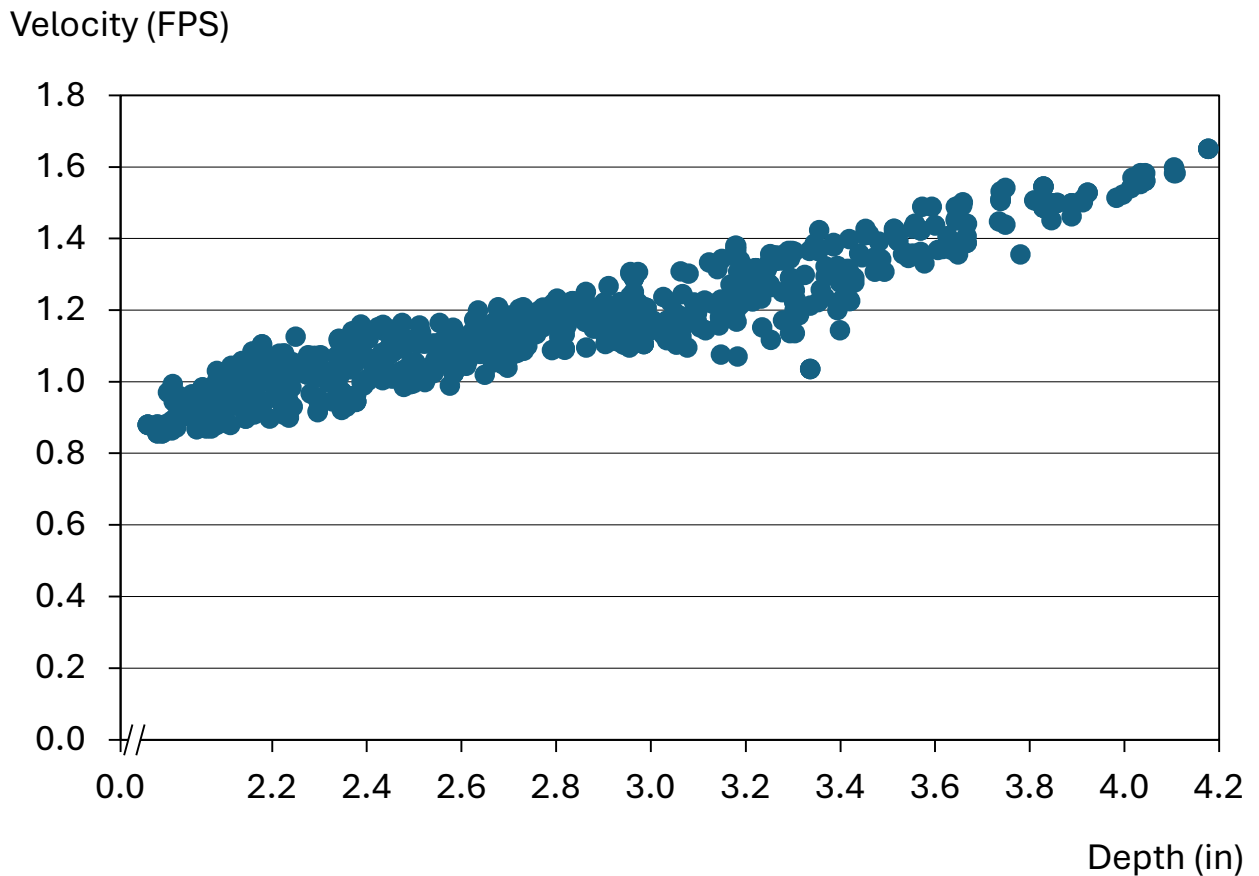
Flow vs. Time

Flow (GPM)



Wet Weather Analysis (8/29, 9/3-9/4)

Velocity vs. Depth



Analysis and notes

Phoenix had rain on three days during the study: 9/19 (.01 inches), 9/26 (1.64 inches) and 9/27 (0.21 inches).

No observed difference between the flow on wet days vs. overall. The average flow is a bit lower but that is a reflection of 1/3 of the rain days being on the weekend and the weekend flow significantly lower than business week days.

Dry Weather Analysis

Overall Data

Flow (GPM)

Max	204.60	Average	86.95
Min	30.51		

Level (in)

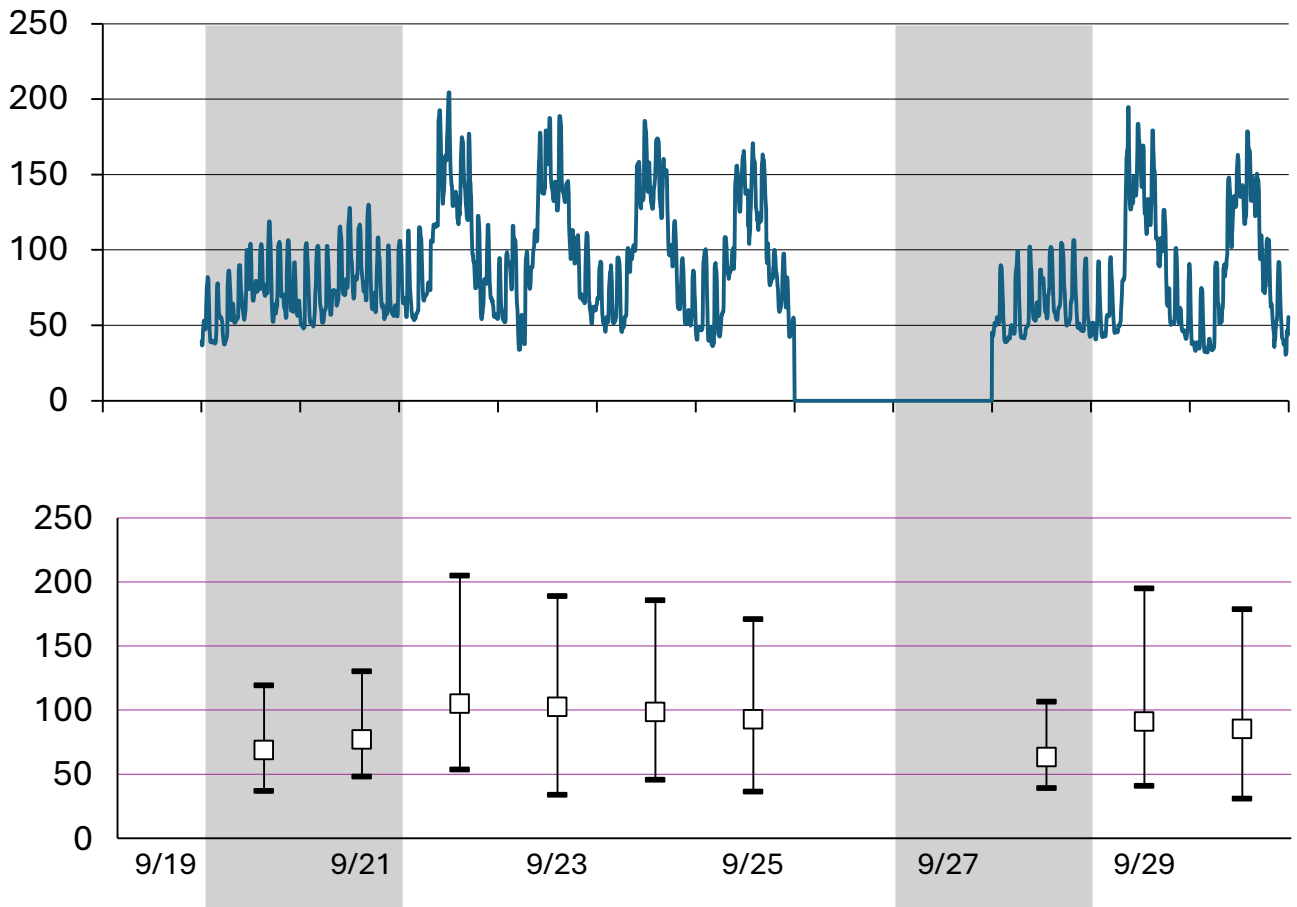
Max	4.35	Average	2.89
Min	1.79		

Velocity (FPS)

Max	1.67	Average	1.16
Min	0.73		

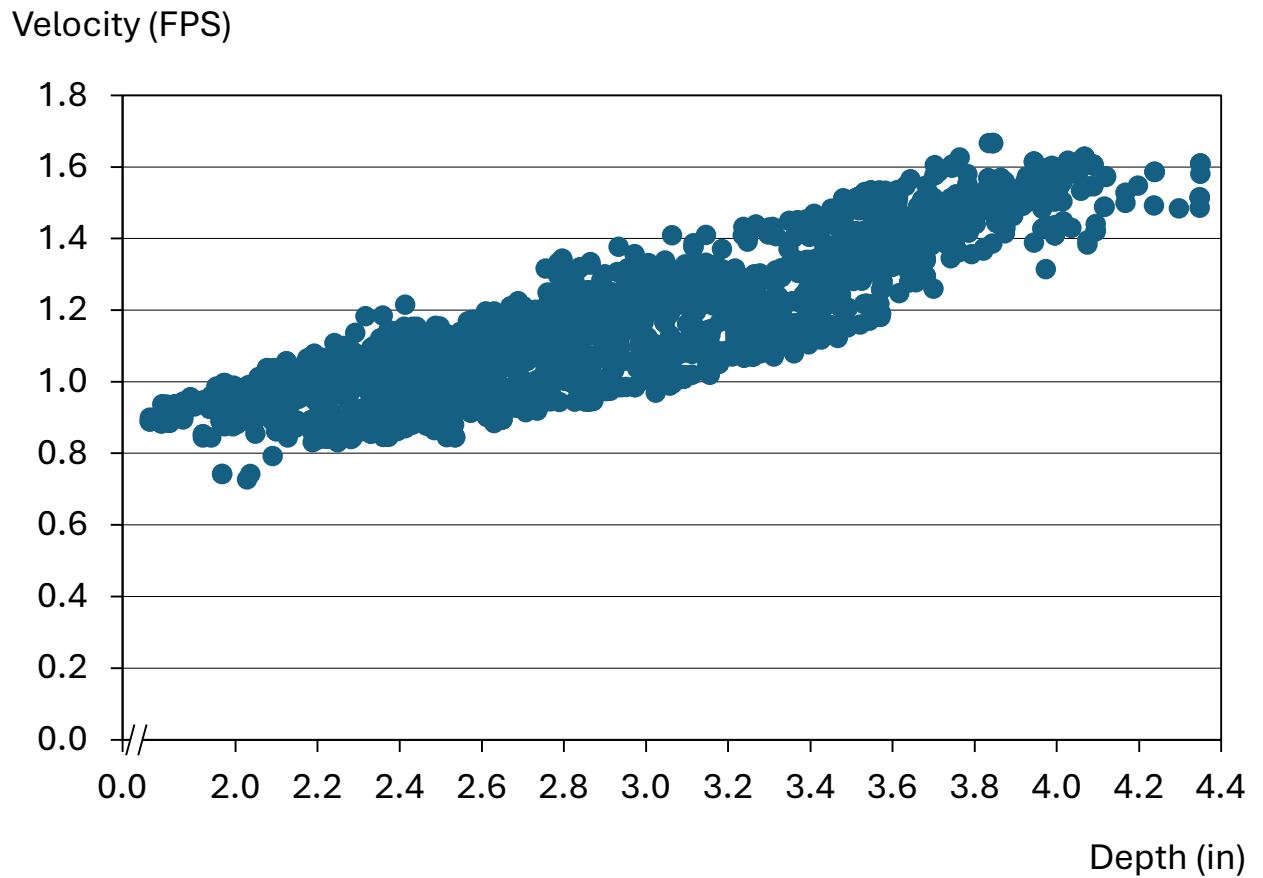
Flow vs. Time

Flow (GPM)



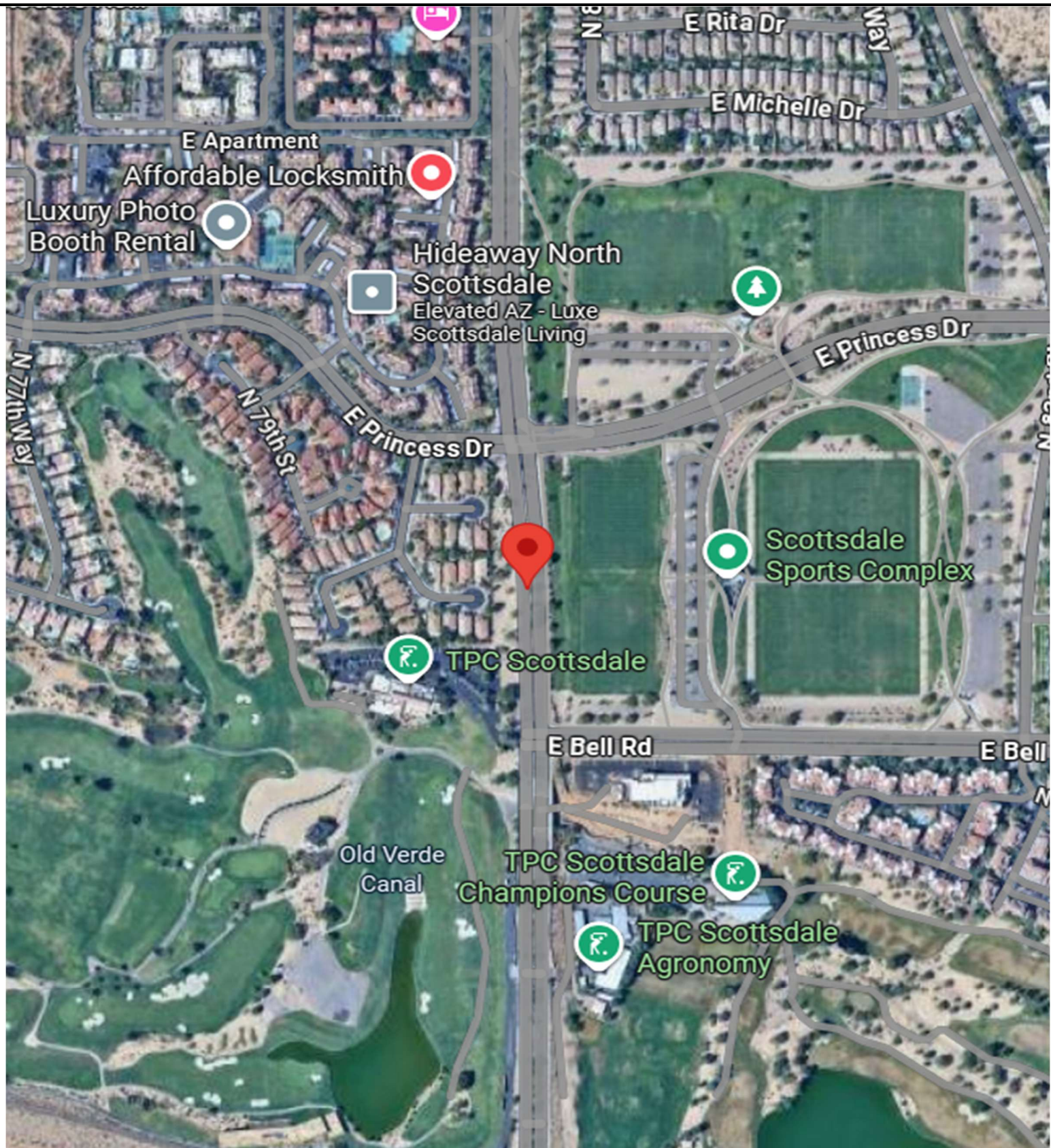
Dry Weather Analysis

Velocity
vs.
Depth



Analysis
and
notes

There is no significant difference in flow on dry days vs. overall



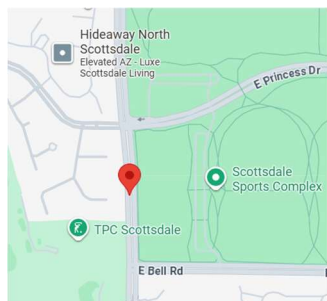
Scottsdale Two Site Flow Study

Manhole: N/A
Pipe Size: 8.25"
Latitude: 33.6416770 N
Longitude: 111.9076023 W

Flow Meter Site Report

Meter Information

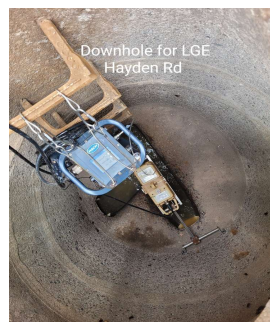
Study Name	Manhole ID	Site Address
LGE Scottsdale	N/A	17020 Hayden Rd, Scottsdale, AZ 85255
Latitude	Longitude	Area Type
33.6416770 N	111.9076023 W	Residential
Installation Date	Installation Crew	Sensor Configuration
9/19/2025	Jordan/Zach/Garrick	Velocity & Ultrasonic
Surcharge Mount	Comments	
No	-	



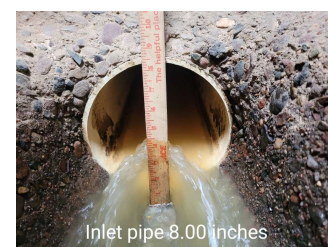
1. Location Map



2. Area Photo



3. Interior MH Photo



4. In Pipe Photo

Manhole Information

Manhole Depth (Feet):	11' 6"
Manhole Material:	PVC

Hydraulics

Surcharge (Feet):	-
Flow Depth at Install (Inches):	2.45
Velocity at Install (FPS):	1.56
Silt Level at Install (Inches):	0
Signs Of Inflow/Infiltration:	No
Overall Hydraulic Rating:	Smooth/Laminar

Pipe Information

Pipe Height (Inches):	8.25
Pipe Width (Inches):	-
Pipe Shape:	Circular
Pipe Material:	Lined
Comments:	n/a

Groundwater Gauge

Existing GWG:	No
Installed New GWG:	No
Type:	-
Groundwater Level	
Distance from Gauge to	
Lowest Pipe Invert (Inches):	-

Other Information

Safety Concerns:	-	Required Traffic Control:	No
Required Police Detail:	No	Cell Reception:	Yes
		Comments:	-

Overall Analysis (Wet and Dry Days)

Overall Data

Flow (GPM)

Max	153.93	Average	71.63
Min	16.44		

Level (in)

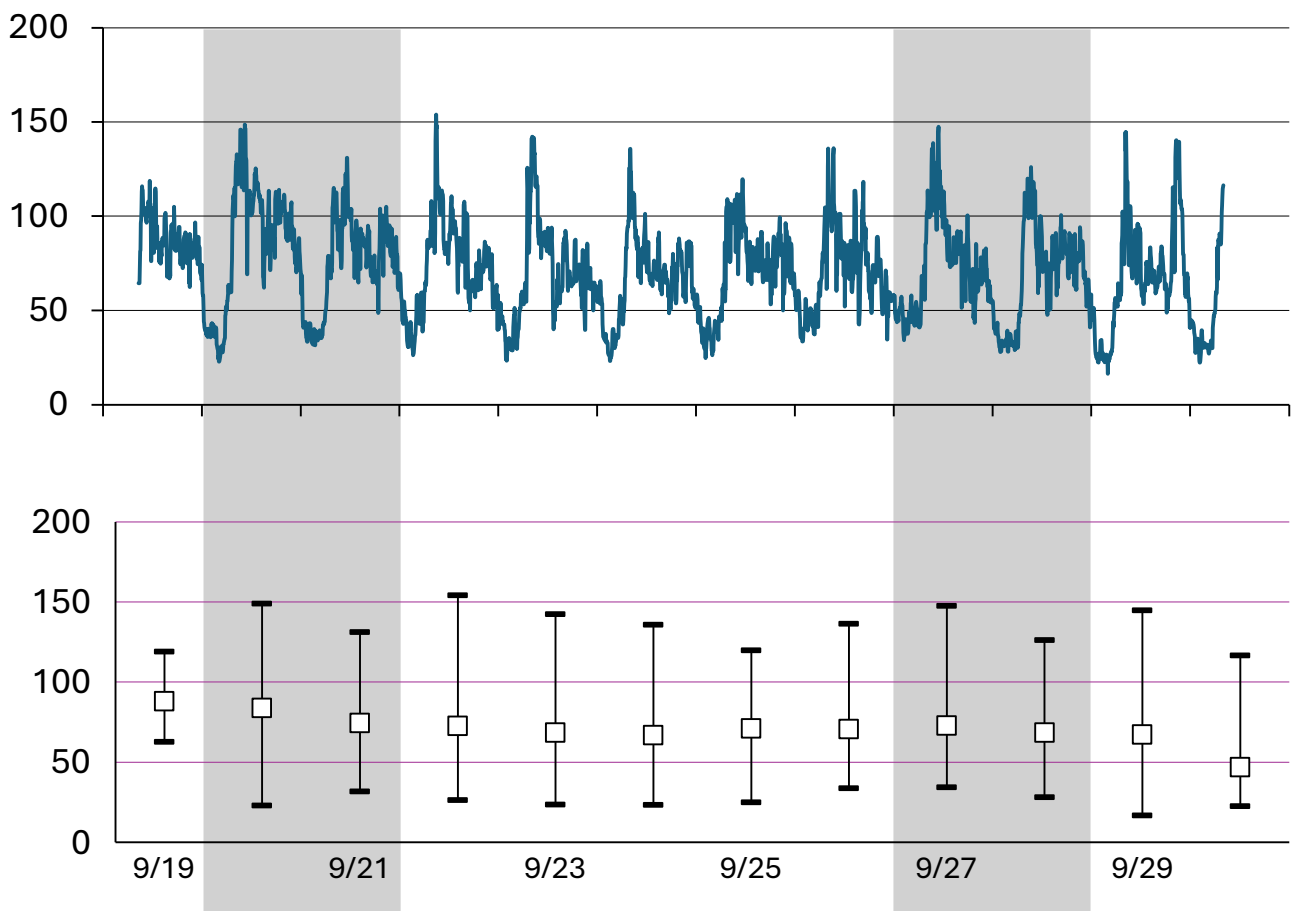
Max	3.78	Average	2.28
Min	1.49		

Velocity (FPS)

Max	2.84	Average	1.90
Min	0.77		

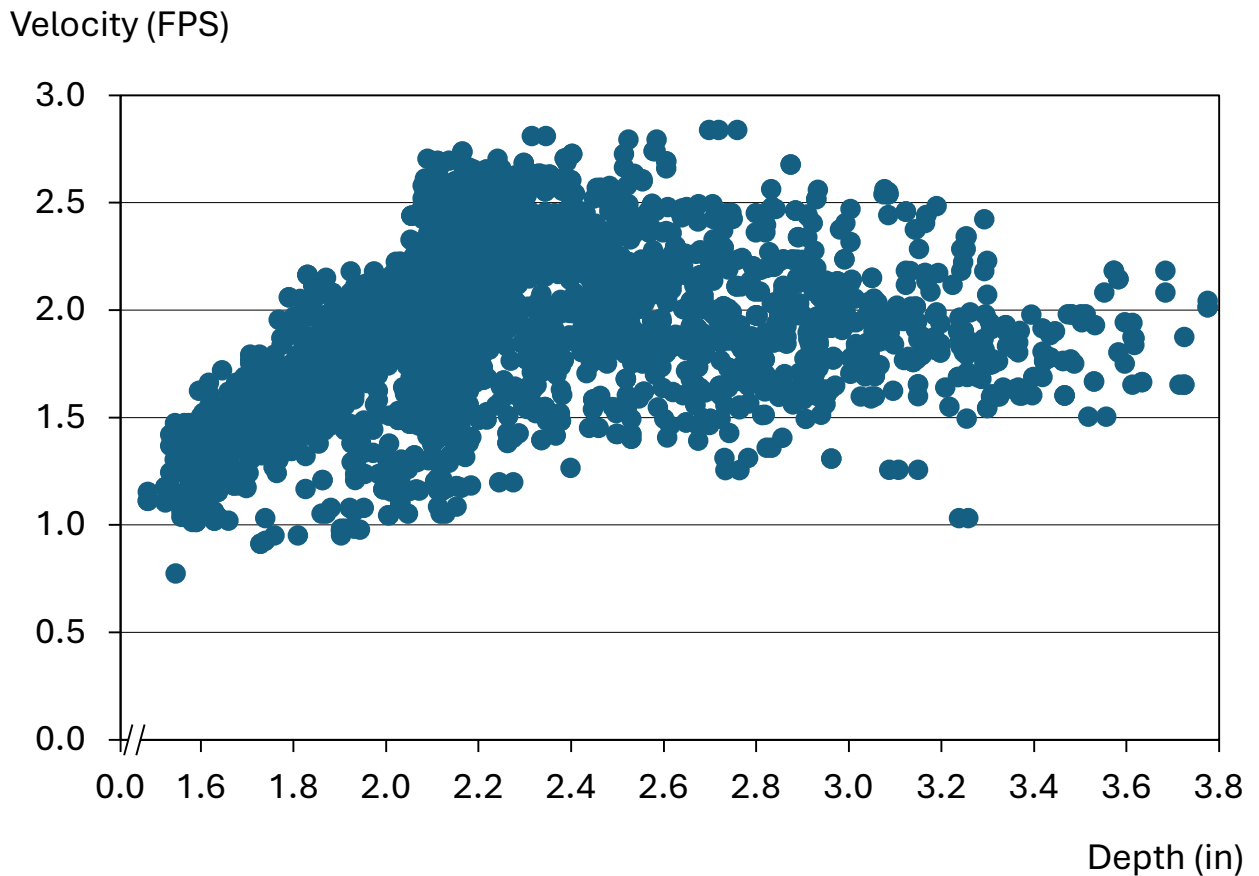
Flow vs. Time

Flow (GPM)



Overall Analysis (Wet and Dry Days)

Velocity
vs.
Depth



Analysis
and
notes

Equipment was installed on 9/19. There were three days of rain (9/19, 9/26-27).

Normal diurnal pattern was observed with no significant change on weekends. However, there is a clear issue once the depth increases to about 2.4 inches and velocity levels out. This is happening mostly during rainy days. While there are many potential reasons for this, we believe the second invert would be causing this issue. During installation, we identified a dry invert with no flow in the manhole. We hypothesize that there are days / times where flow comes from this invert (especially on rain days) which raises the overall level of flow and potentially increases the water turbulence reducing the overall velocity.

We used the data of level, flow, and velocity to fit a slope and roughness coefficient consistent. Based off of these input variables, we estimate total maximum flow to be 377 GPM with an 80% maximum to be 301 GPM.

Wet Weather Analysis (9/19, 9/26-9/27)

Overall Data

Flow (GPM)

Max	147.38	Average	75.47
Min	33.51		

Level (in)

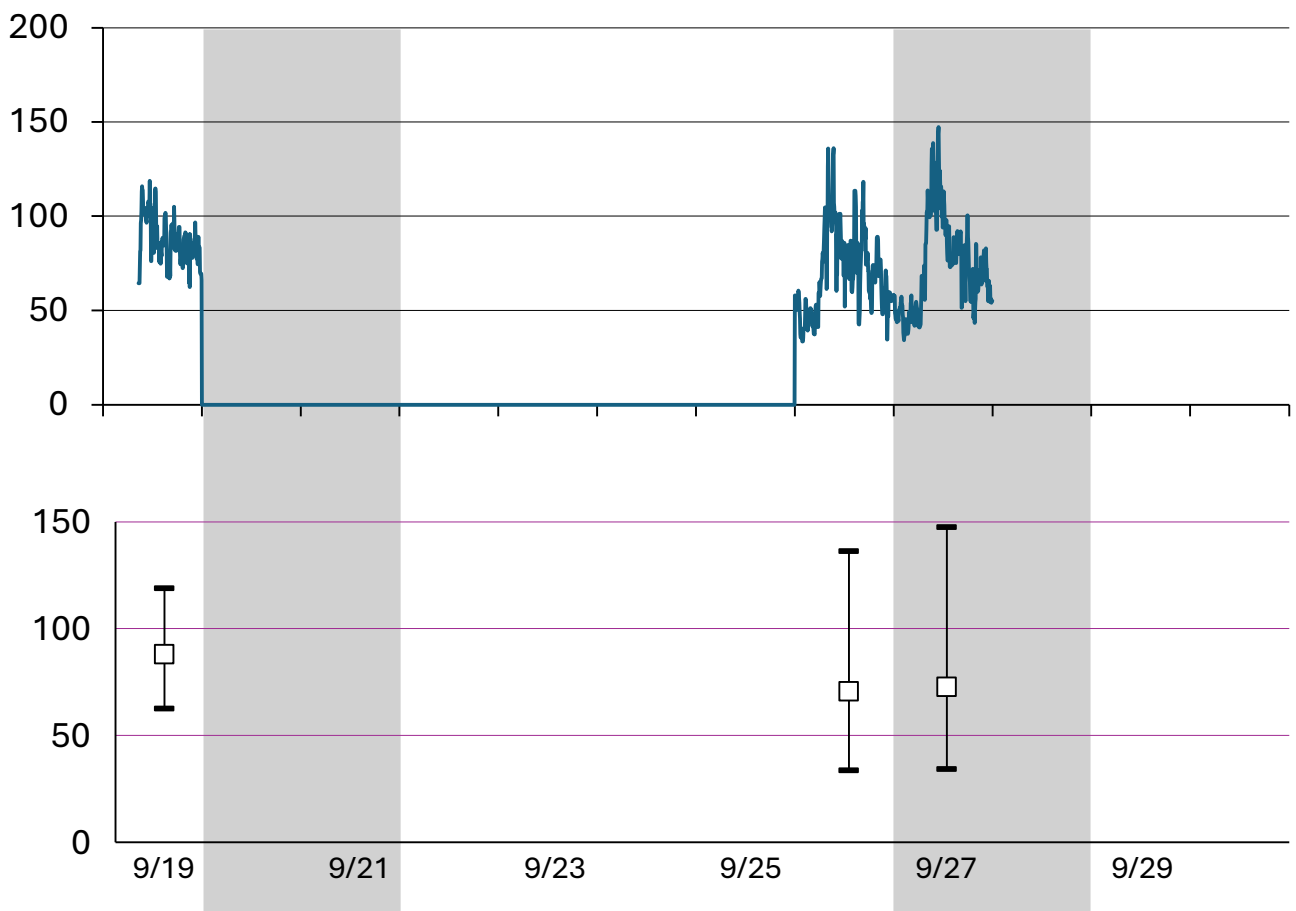
Max	3.47	Average	2.37
Min	1.68		

Velocity (FPS)

Max	2.84	Average	1.93
Min	1.03		

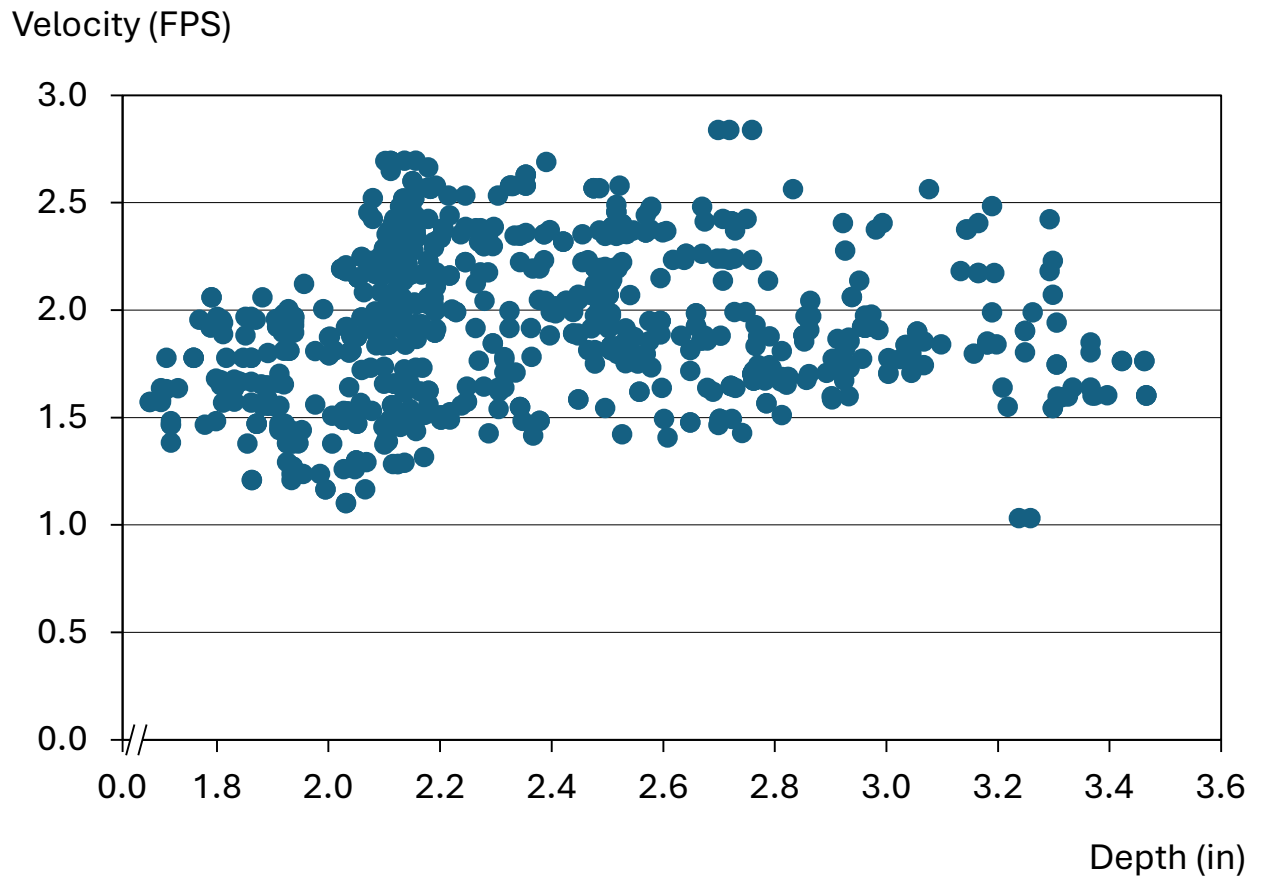
Flow vs. Time

Flow (GPM)



Wet Weather Analysis (8/29, 9/3-9/4)

Velocity
vs.
Depth



Analysis
and
notes

Phoenix had rain on three days during the study: 9/19 (.01 inches), 9/26 (1.64 inches) and 9/27 (0.21 inches).

As discussed in the overall analysis, we typically see a normal linear relationship between velocity and depth even during wet days. The pattern we see in this study is different: velocity relatively uncorrelated with depth on wet days. We believe the second invert has additional flow from somewhere upstream when it is raining.

Dry Weather Analysis

Overall Data

Flow (GPM)

Max	153.93	Average	70.42
Min	16.44		

Level (in)

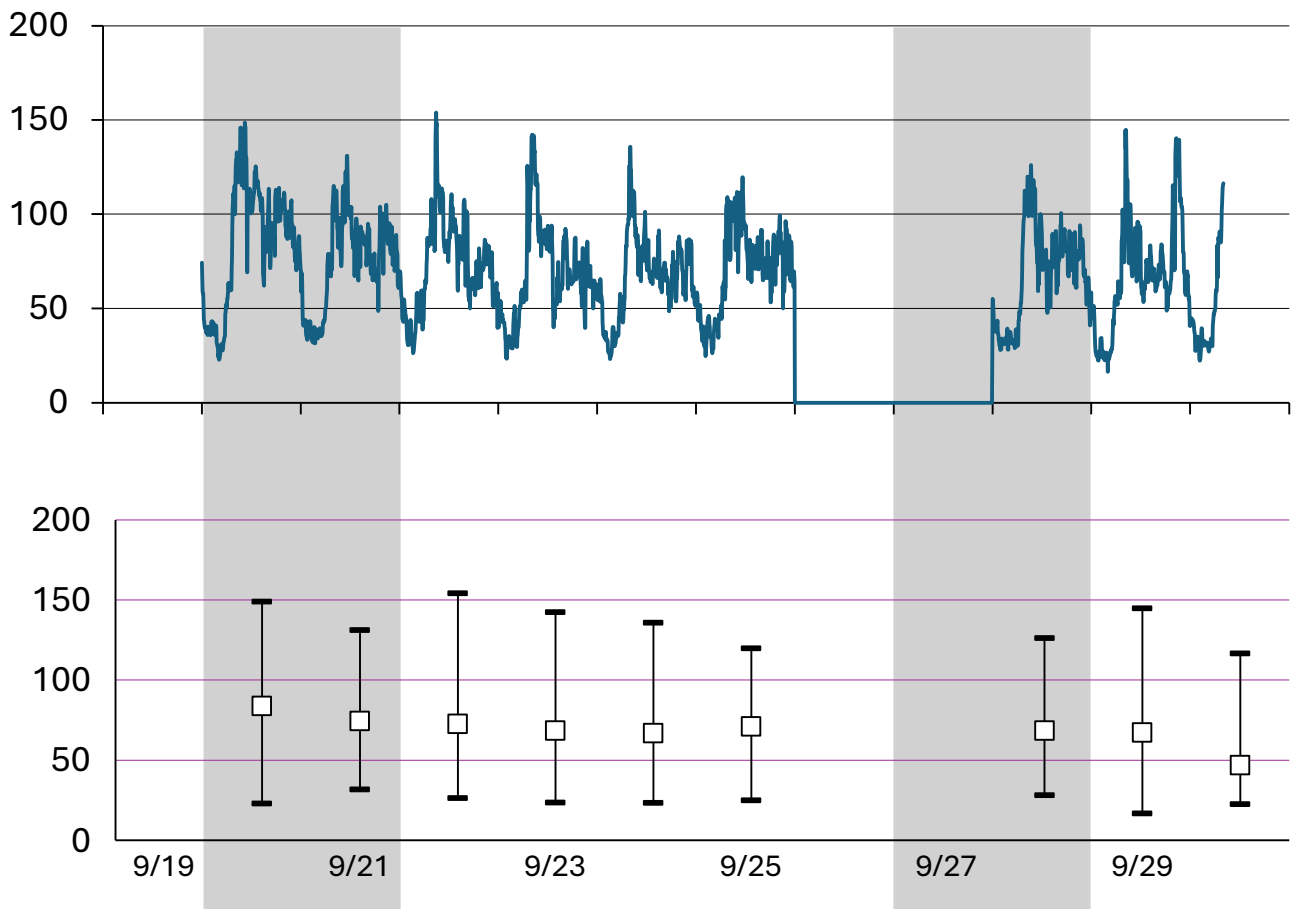
Max	3.78	Average	2.25
Min	1.49		

Velocity (FPS)

Max	2.81	Average	1.90
Min	0.77		

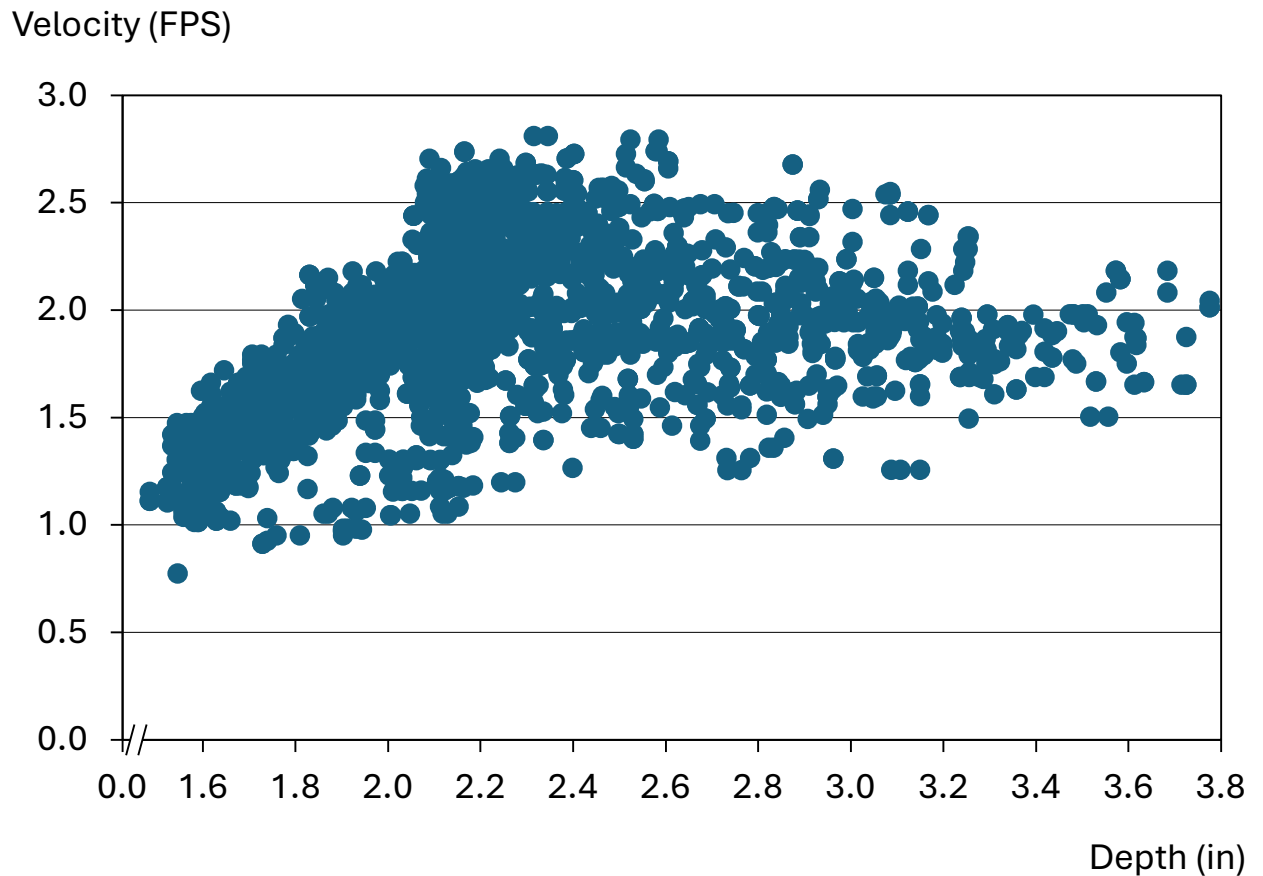
Flow vs. Time

Flow (GPM)



Dry Weather Analysis

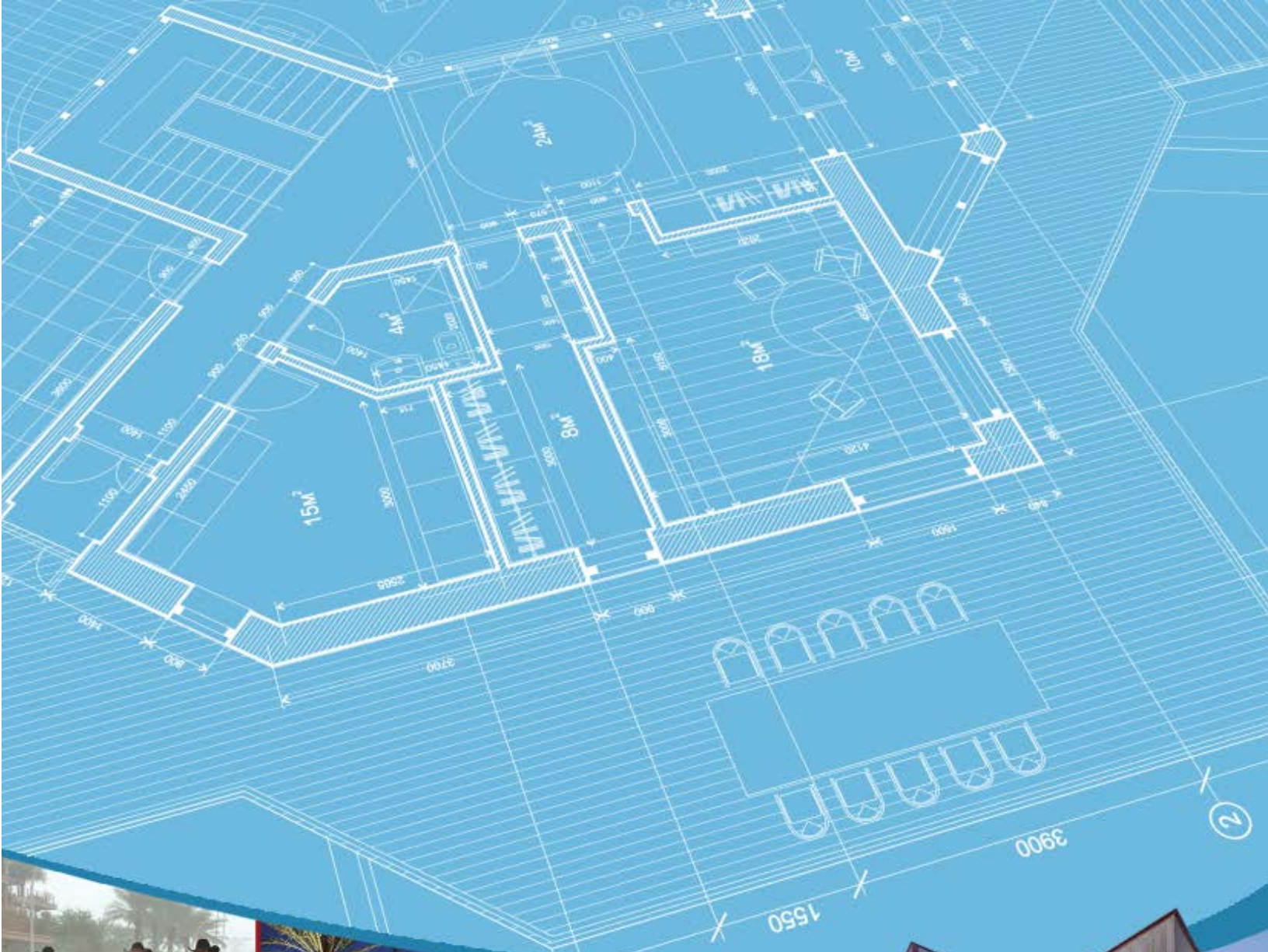
Velocity
vs.
Depth



Analysis
and
notes

We continue to see velocity become uncorrelated with depth after about 2.4 inches even on dry days.

There is no significant difference in flow on dry days vs. overall



DESIGN STANDARDS & POLICIES MANUAL

LAND USE	DEMAND (gpd)	DESIGN PEAKING FACTOR
<i>Commercial/Retail</i>	0.5 per sq. ft.	3
<i>Office</i>	0.4 per sq. ft.	3
<i>Restaurant</i>	1.2 per sq. ft.	6
<i>High Density Condominium (Condo)</i>	140 per unit	4.5
<i>Resort Hotel (includes site amenities)</i>	380 per room.	4.5
<i>School: without cafeteria</i>	30 per student	6
<i>School: with cafeteria</i>	50 per student	6
<i>Cultural</i>	0.1 per sq. ft.	3
<i>Clubhouse for Subdivision</i>	100 per patron x 2	4.5
<i>Golf Course</i>	patrons per du per day	
<i>Fitness Center/ Spa/ Health club</i>	0.8 per sq. ft.	3.5

FIGURE 7-1.2 AVERAGE DAY SEWER DEMAND IN GALLONS PER DAY & PEAKING FACTORS BY LAND USE

HYDRAULIC DESIGN

No public SS lines will be less than 8 inches in diameter unless permission is received in writing from the Water Resources Department.

SS lines shall be designed and constructed to give mean full flow velocities equal to or greater than 2.5 fps, based upon Manning's Formula, using an "n" value of 0.013.

To prevent abrasion and erosion of the pipe material, the maximum velocity will be limited to 10 fps at estimated peak flow. Where velocities exceed this maximum figure, submit a hydraulic analysis along with construction recommendations to the Water Resources Department for consideration. In no case will velocities greater than 15 fps be allowed.

Actual velocities shall be analyzed for minimum, average day and peak day design flow conditions for each reach of pipe.

The SS system shall be designed to achieve uniform flow velocities through consistent slopes. Abrupt changes in slope shall be evaluated for hydraulic jump.

The depth to diameter ratio (d/D) for gravity SS pipes 12 inches in diameter and less shall not exceed 0.65 in the ultimate peak flow condition. This d/D ratio includes an allowance for system infiltration and inflow.

The d/D for gravity drains greater than 12 inches diameter shall not exceed 0.70 for the ultimate peak flow condition. This d/D includes an allowance for system infiltration and inflow.

Measures to mitigate hydrogen sulfide shall be analyzed at manhole drops, abrupt changes in pipe slope or direction and at changes in pipe diameter.

MANHOLES AND CLEAN OUTS

Manholes in city streets shall be located near the center of the inside traffic lane, rather than on or near the line separating traffic lanes. Manholes shall not be in bike trails, equestrian trails, sidewalks, crosswalks or wash crossings. Manholes are required at all

7-1.404

7-1.405



**City of Phoenix
Water Services Department**

**DESIGN STANDARDS MANUAL FOR
WATER AND WASTEWATER SYSTEMS**

2021

**Water Services Department
200 West Washington Street
Phoenix, Arizona 85003-1697
Phone: (602) 495-5601
Fax: (602) 495-5461**

in Chapter IV, Section C), are not always adequate to meet water demands. For some projects, a detailed analysis of domestic and fire flow demands may be required to properly define requirements for system design.

1. Water and Sewer Design Flows

The following **Table 8, Water and Wastewater Design Flows** shall be used to calculate both water and sewer design flows utilized in the preparation of engineering design reports, plans, and specifications.

Table 8. Water and Wastewater Design Flows.

Land Use	Unit	Water Average Daily Flow/Unit (gal)	Wastewater Average Daily flow/Unit (gal)
Single Family Residential	Dwelling	360	240
Multi-family	Dwelling	240	180
Commercial (retail/mall)	1000 ft ²	125	75
Commercial (office)	1000 ft ²	115	90
Warehousing/Big Box Retail	1000 ft ²	30	25
Industrial	1000 ft ²	65	50
Schools	Student	25	20
Hotel (no restaurant)	Room	140	100
Hotel (with restaurant)	Room	200	150
Resort	Room	300	210
Hospital (all flows)	Bed	500	300
Landscape Water Requirements			
General Landscaping	Acre	4,374	N/A
Public Right of Way or Streetscape	Acre	1,339	N/A
Surface Water	Acre	5,335	N/A

NOTES: The following italicized notes are for Table 8, Water and Wastewater Design Flows

Complete design flows are not provided for ***industrial and hospital facilities*** because case-by-case evaluation is necessary due to varying water demands observed for these use types. Some industrial uses such as data warehouses, food processing, bottling plants, and semi-conductor manufacturing can use more than ten times as much water as compared to warehousing or dry assembly manufacturing with no cooling tower use. Water use in hospitals varies greatly depending upon cooling tower and boiler use, the extent to which the hospital is used as a research and teaching facility, the amount of out-patient versus in-patient services provided, and the types of equipment used. Estimates of anticipated water use and wastewater generation must be produced for each new development or major expansion using projections of demands taking into account the following types of categories:

- ***Water for cooling towers:*** Cooling towers use can make up more than fifty percent of water demand at industrial facilities having large refrigeration units or cooling of servers. In most cases, cooling towers use twenty to forty percent of the water requirements for industrial operations and hospitals.
- ***Water used as an input for production:*** In some manufacturing operations, water is used as an input in the manufacturing process and must be included in demand projections because of the large volumes used. Examples include ice-making, soft-drink or water bottling operations, and food manufacturing such as industrial bakeries.
- ***Water used in production/activities:*** In many manufacturing operations water is used for cooling, cleaning, or other operational activities and must be included in demand projections. Examples include metal forming and finishing, semi-conductor wafer production, and aerospace parts manufacturing. Processes employing newer technologies tend to use less water than older technologies, but estimates must be made on a location and process-specific basis. Some medical facilities are now using the newer medical imaging techniques and sterilization processes that use little or no water, while some medical equipment still requires significant amounts of water.
- ***Bed to space ratios and mix of services:*** Bed to space ratios and services provided in hospitals can vary greatly. These variations depend upon the proportion of space necessary to provide 24/7 nursing care, full linen service, and full food service

to patients staying overnight. Furthermore, some hospitals are highly specialized and focus on particular types of treatment and/or research while others provide general and emergency services only. Water use on a per-square-foot or per-bed-basis can even vary significantly between different parts of hospitals, so large expansions will require an individual analysis.

2. Water Peak Flow

Peak Flow shall be calculated as 1.7 times the average daily flow.

NOTE: For clarification, the following example characterizes the calculations performed to determine the design flows and quantities involved in a hypothetical facility.

EXAMPLE: Hypothetical water demand/flow evaluation (not including fire flows).

ASSUME: A 1000 dwelling unit multi-family development.

CRITERIA: From **Table 8, Water and Wastewater Design Flows.**

Average daily flow = 240 gallons per unit per day (gpud)

Average total daily flow = 1,000 x 240 = 240,000 gallons per day (GPD)

Peak daily flow = 240,000 GPD x 1.7 (peaking factor)

Peak daily flow = 408,000 GPD

3. Sewer Peak Flow

All gravity sewer mains shall be designed for peak flow conditions. Peak flow is calculated as the product of the peaking factor and the average daily flow. The peaking factor should be calculated from Harmon's formula.

Design Flow = Peak Flow = Q Peak = Q avg [1+14/ (4+ P^{1/2})], Where P = Population/1,000

F. WATER AND SEWER MAIN ABANDONMENT METHODS

There are three approved methods of abandoning water and sewer mains in public ROW and easements:

- a. Total removal of pipe.
- b. Crush pipe in place by mechanical means. This cannot be applied to asbestos cement pipe.
- c. Leave pipe in place and fill with low strength grout.

No other methods are acceptable.

G. WATER AND SEWER STUBS OR TAPS AHEAD OF PAVING

City of Phoenix does not allow new stubs or taps ahead of paving unless the property owner can provide a conceptual design report and a site plan demonstrating the appropriate sizing and location of the mains or stubs. This applies to connections such as water/sewer stubs, water/sewer mains and service taps for fire lines and/or domestic use. The request for taps ahead of paving shall be submitted by the developer through a Water and Sewer Technical Appeal.

If the City approves the request for taps ahead of paving, and the size or location changes after the installation due to design changes, or for any other reason, it shall be the property owner's responsibility to abandon any unused infrastructure at the property owner's expense.

H. CROSS CONNECTIONS AND BACKFLOW PREVENTION

1. Cross Connection